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Abuse Elevation Control Mechanism: Bypass User Account Control

Other sub-techniques of Abuse Elevation Control Mechanism (6)

Adversaries may bypass UAC mechanisms to elevate process privileges on system. Windows User Account Control (UAC) allows a program to elevate its privileges (tracked as integrity levels ranging from low to high) to perform a task under administrator-level permissions, possibly by prompting the user for confirmation. The impact to the user ranges from denying the operation under high enforcement to allowing the user to perform the action if they are in the local administrators group and click through the prompt or allowing them to enter an administrator password to complete the action.^[1]

If the UAC protection level of a computer is set to anything but the highest level, certain Windows programs can elevate privileges or execute some elevated Component Object Model objects without prompting the user through the UAC notification box.^{[2][3]} An example of this is use of Rundll32 to load a specifically crafted DLL which loads an auto-elevated Component Object Model object and performs a file operation in a protected directory which would typically require elevated access. Malicious software may also be injected into a trusted process to gain elevated privileges without prompting a user.^[4]

Many methods have been discovered to bypass UAC. The Github readme page for UACME contains an extensive list of methods^[5] that have been discovered and implemented, but may not be a comprehensive list of bypasses. Additional bypass methods are regularly discovered and some used in the wild, such as:

- `eventvwr.exe` can auto-elevate and execute a specified binary or script.^{[6][7]}

Another bypass is possible through some lateral movement techniques if credentials for an account with administrator privileges are known, since UAC is a single system security mechanism, and the privilege or integrity of a process running on one system will be unknown on remote systems and default to high integrity.^[8]

Procedure Examples

ID	Name	Description
S0584	AppleJeus	AppleJeus has presented the user with a UAC prompt to elevate privileges while installing. ^[9]
G0016	APT29	APT29 has bypassed UAC. ^[10]
G0067	APT37	APT37 has a function in the initial dropper to bypass Windows UAC in order to execute the next payload with higher privileges. ^[11]
G0082	APT38	APT38 has used the legitimate application <code>ieinstal.exe</code> to bypass UAC. ^[12]
S0129	Autolt backdoor	Autolt backdoor attempts to escalate privileges by bypassing User Access Control. ^[13]
S0640	Avaddon	Avaddon bypasses UAC using the CMSTPLUA COM interface. ^[14]
S0606	Bad Rabbit	Bad Rabbit has attempted to bypass UAC and gain elevated administrative privileges. ^[15]
S1081	BADHATCH	BADHATCH can utilize the CMSTPLUA COM interface and the SilentCleanup task to bypass UAC. ^[16]
S0570	BitPaymer	BitPaymer can suppress UAC prompts by setting the <code>HKCU\Software\Classes\ms-settings\shell\open\command</code> registry key on Windows 10 or <code>HKCU\Software\Classes\mscfile\shell\open\command</code> on Windows 7 and launching the <code>eventvwr.msc</code> process, which launches BitPaymer with elevated privileges. ^[17]
S1068	BlackCat	BlackCat can bypass UAC to escalate privileges. ^[18]
S0089	BlackEnergy	BlackEnergy attempts to bypass default User Access Control (UAC) settings by exploiting a backward-compatibility setting found in Windows 7 and later. ^[19]
G0060	BRONZE BUTLER	BRONZE BUTLER has used a Windows 10 specific tool and xxmm to bypass UAC for privilege escalation. ^{[20][21]}
S1039	Bumblebee	Bumblebee has the ability to bypass UAC to deploy post exploitation tools with elevated privileges. ^[22]
S1149	CHIMNEYSWEEP	CHIMNEYSWEEP can make use of the Windows <code>SilentCleanup</code> scheduled task to execute its payload with elevated privileges. ^[23]
S0660	Clambling	Clambling has the ability to bypass UAC using a <code>passuac.dll</code> file. ^{[24][25]}
G0080	Cobalt Group	Cobalt Group has bypassed UAC. ^[26]
S0154	Cobalt Strike	Cobalt Strike can use a number of known techniques to bypass Windows UAC. ^{[27][28]}
S0527	CSPY Downloader	CSPY Downloader can bypass UAC using the SilentCleanup task to execute the binary with elevated privileges. ^[29]
S1111	DarkGate	DarkGate uses two distinct User Account Control (UAC) bypass techniques to escalate privileges. ^[30]
S0134	Downdelph	Downdelph bypasses UAC to escalate privileges by using a custom "RedirectEXE" shim database. ^[31]
G1006	Earth Lusca	Earth Lusca has used the Fodhelper UAC bypass technique to gain elevated privileges. ^[32]
S0363	Empire	Empire includes various modules to attempt to bypass UAC for escalation of privileges. ^[33]
G0120	Evilnum	Evilnum has used PowerShell to bypass UAC. ^[34]
S0182	FinFisher	FinFisher performs UAC bypass. ^{[35][36]}
S0666	Gelsemium	Gelsemium can bypass UAC to elevate process privileges on a compromised host. ^[37]
S0531	Grandoreiro	Grandoreiro can bypass UAC by registering as the default handler for .MSC files. ^[38]
S0132	H1N1	H1N1 bypasses user access control by using a DLL hijacking vulnerability in the Windows Update Standalone Installer (<code>wusa.exe</code>). ^[39]
S9007	HTTPTroy	HTTPTroy has leveraged the ability to execute commands with system privileges using the <code>run <EXECUTABLE> <ARGS></code> command. ^[40]
S0260	InvisiMole	InvisiMole can use fileless UAC bypass and create an elevated COM object to escalate privileges. ^{[41][42]}
S0250	Koadic	Koadic has 2 methods for elevating integrity. It can bypass UAC through <code>eventvwr.exe</code> and <code>adclt.exe</code> . ^[43]
S0669	KOCTOPUS	KOCTOPUS will perform UAC bypass either through <code>fodhelper.exe</code> or <code>eventvwr.exe</code> . ^[44]

ID: T1548.002

Sub-technique of: T1548

ⓘ [Tactic: Privilege Escalation](#)

ⓘ [Platforms: Windows](#)

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	S0356	KONNI	KONNI has bypassed UAC by performing token impersonation as well as an RPC-based method, this included bypassing UAC set to "AlwaysNotify". ^{[45][46]}
	S1199	LockBit 2.0	LockBit 2.0 can bypass UAC through creating the Registry key <code>HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Microsoft\Windows NT\CurrentVersion\ICM\Calibration</code> . ^{[47][48]}
	S1202	LockBit 3.0	LockBit 3.0 can bypass UAC to execute code with elevated privileges through an elevated Component Object Model (COM) interface. ^[49]
	S0447	Lokibot	Lokibot has utilized multiple techniques to bypass UAC. ^[50]
	G1051	Medusa Group	Medusa Group has attempted to bypass UAC using Component Object Model (COM) interface. ^[51]
	G0069	MuddyWater	MuddyWater uses various techniques to bypass UAC. ^{[52][53]}
	C0006	Operation Honeybee	During Operation Honeybee, the threat actors used the malicious NTWDBLIB.DLL and <code>clliconfig.exe</code> to bypass UAC protections. ^[54]
	G0040	Patchwork	Patchwork bypassed User Access Control (UAC). ^[55]
	S0501	PipeMon	PipeMon installer can use UAC bypass techniques to install the payload. ^[56]
	S0254	PLAINTEE	An older variant of PLAINTEE performs UAC bypass. ^[57]
	S0378	PoshC2	PoshC2 can utilize multiple methods to bypass UAC. ^[58]
	S0192	Pupy	Pupy can bypass Windows UAC through either DLL hijacking, <code>eventvwr</code> , or <code>appPaths</code> . ^[59]
	S1242	Qilin	Qilin can bypass standard user access controls by using stolen tokens to launch processes at an elevated security context. ^[60]
	S0262	QuasarRAT	QuasarRAT can generate a UAC pop-up Window to prompt the target user to run a command as the administrator. ^[61]
	S0458	Ramsay	Ramsay can use UACMe for privilege escalation. ^{[62][63]}
	S1130	Raspberry Robin	Raspberry Robin will use the legitimate Windows utility <code>fodhelper.exe</code> to run processes at elevated privileges without requiring a User Account Control prompt. ^[64]
	S0662	RCSession	RCSession can bypass UAC to escalate privileges. ^[24]
	S0332	Remcos	Remcos has a command for UAC bypassing. ^[65]
	S0148	RTM	RTM can attempt to run the program as admin, then show a fake error message and a legitimate UAC bypass prompt to the user in an attempt to socially engineer the user into escalating privileges. ^[66]
	S1018	Saint Bot	Saint Bot has attempted to bypass UAC using <code>fodhelper.exe</code> to escalate privileges. ^[67]
	S0074	Sakula	Sakula contains UAC bypass code for both 32- and 64-bit systems. ^[68]
	S0140	Shamoon	Shamoon attempts to disable UAC remote restrictions by modifying the Registry. ^[69]
	S0444	ShimRat	ShimRat has hijacked the <code>cryptbase.dll</code> within <code>migwiz.exe</code> to escalate privileges. This prevented the User Access Control window from appearing. ^[70]
	S0692	SILENTRINITY	SILENTRINITY contains a number of modules that can bypass UAC, including through Windows's Device Manager, Manage Optional Features, and an image hijack on the <code>.msc</code> file extension. ^[71]
	S0633	Sliver	Sliver can leverage multiple techniques to bypass User Account Control (UAC) on Windows systems. ^[72]
	G0027	Threat Group-3390	A Threat Group-3390 tool can use a public UAC bypass method to elevate privileges. ^[73]
	S0116	UACMe	UACMe contains many methods for bypassing Windows User Account Control on multiple versions of the operating system. ^[5]
	S0275	UPPERCUT	UPPERCUT contains functionality to bypass UAC. ^{[74][75]}
	S0670	WarzoneRAT	WarzoneRAT can use <code>sdclt.exe</code> to bypass UAC in Windows 10 to escalate privileges; for older Windows versions WarzoneRAT can use the <code>IFileOperation</code> exploit to bypass the UAC module. ^{[76][77]}
	S0612	WastedLocker	WastedLocker can perform a UAC bypass if it is not executed with administrator rights or if the infected host runs Windows Vista or later. ^[78]
	S0141	Winnti for Windows	Winnti for Windows can use a variant of the <code>sysprep</code> UAC bypass. ^[79]
	S0230	ZeroT	Many ZeroT samples can perform UAC bypass by using <code>eventvwr.exe</code> to execute a malicious file. ^[80]

Mitigations

ID	Mitigation	Description
M1047	Audit	Check for common UAC bypass weaknesses on Windows systems to be aware of the risk posture and address issues where appropriate. ^[5]
M1026	Privileged Account Management	Remove users from the local administrator group on systems.
M1051	Update Software	Consider updating Windows to the latest version and patch level to utilize the latest protective measures against UAC bypass. ^[5]
M1052	User Account Control	Although UAC bypass techniques exist, it is still prudent to use the highest enforcement level for UAC when possible and mitigate bypass opportunities that exist with techniques such as DLL.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0388	Detection Strategy for T1548.002 – Bypass User Account Control (UAC)	AN1094	Detects a multi-event behavior chain involving UAC bypass attempts via known auto-elevated binaries (e.g., <code>eventvwr.exe</code> , <code>sdclt.exe</code>), unauthorized Registry changes to UAC-related keys, and anomalous process execution with elevated privileges but lacking standard parent-child lineage. Suspicious patterns include invocation of auto-elevated COM objects or manipulation of isolatedCommand Registry entries without consent prompts.

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